



Class 9

MATTER IN OUR SURROUNDINGS

Ch 1 | Class 9th (Science)

Handwritten Notes



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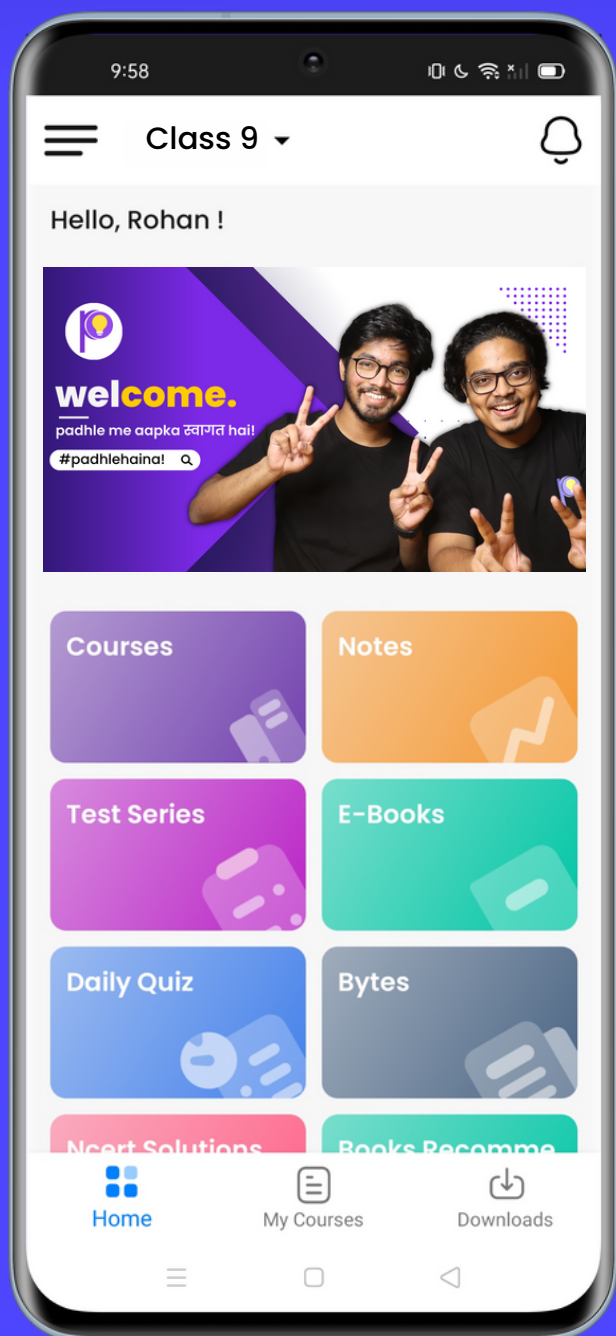
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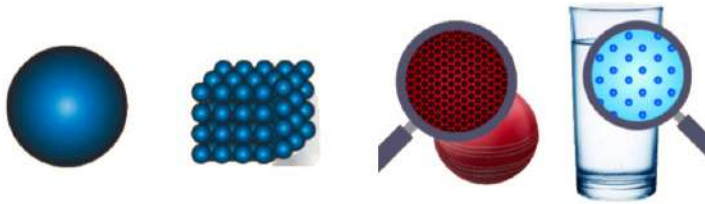


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MATTER IN OUR SURROUNDING

* What is Matter?

- Matter is made up of small particles.
- We can either see it, touch it, feel it or taste it, or all.



- For a long time, two schools of thought prevailed regarding the nature of matter.
 - One school believed matter to be continuous like a block of wood, whereas, the other thought that matter was made up of particles like sand.

* What are characteristics of particles matter

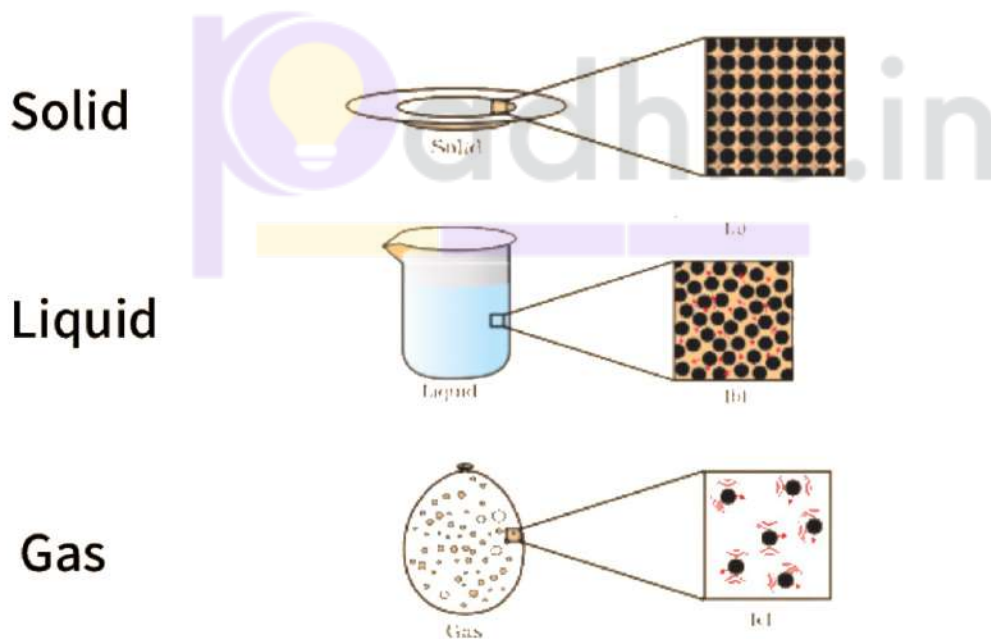
1. Particles of Matter have spaces between them.
2. Particles of Matter are continuously moving:
 - Particles of matter are continuously moving, that is, they possess what we call the kinetic energy.
 - As the temperature rises, particles move faster.
 - So, we can say that with increase in temperature the kinetic energy of the particles also increases.

- Particles of matter intermix on their own with each other.
- They do so by getting into the spaces between the particles.
- This intermixing of particles of two different types of matter on their own is called diffusion.

3. Particles of Matter attract each other:

- This force keeps the particles together.
- The strength of this force of attraction varies from one kind of matter to another.

* What are States of Matter ?



	<u>Shape</u>	<u>Volume</u>
Solid	Definite	Definite
Liquid	Indefinite	Indefinite
Gas	Indefinite	Indefinite

	Intermolecular Force of Attraction	Intermolecular Spaces
Solid	Maximum	Minimum
Liquid	Moderate	Moderate
Gas	Minimum	Maximum

Solid :

- Have a definite shape, distinct boundaries and fixed volumes, that is, have negligible compressibility.
- Solids have a tendency to maintain their shape when subjected to outside force.
- Solids may break under force but it is difficult to change their shape, so they are rigid.

- (a) What about a rubber band, can it change its shape on stretching? Is it a solid?
- (b) What about sugar and salt? When kept in different jars these take the shape of the jar. Are they solid?
- (c) What about a sponge? It is a solid yet we are able to compress it. Why?

Liquids :

- Have a definite shape, distinct boundaries and fixed volume, that is, have negligible compressibility.
- Solids have a tendency to maintain their shape when subject to outside force.
- Solids may break under force but it is difficult to change their shape, so they are rigid.
- They take up the shape of the container in which they are kept.
 - Liquids flow and change shape, so they are not rigid but can be called fluid.
- The gases from the atmosphere diffuse and dissolve in water.
- These gases, especially oxygen and carbon dioxide, are essential for the survival of aquatic animals and plants.
- The rate of diffusion of liquids is higher than that of solids.
- This is due to the fact that in the liquid state, particles move freely and have greater space between each other as compared to particles in the solid state.

Gases :

- They take up the shape of the container in which they are kept.
 - Liquids flow and change shape, so they are not rigid but can be called fluid.
- The gases from the atmosphere diffuse and dissolve in water.
- These gases, especially oxygen and carbon dioxide, are essential for the survival of aquatic animals and plants.

* Plasma

- Plasma is superheated matter - so hot that the electrons are ripped away from the atoms forming an ionised gas.

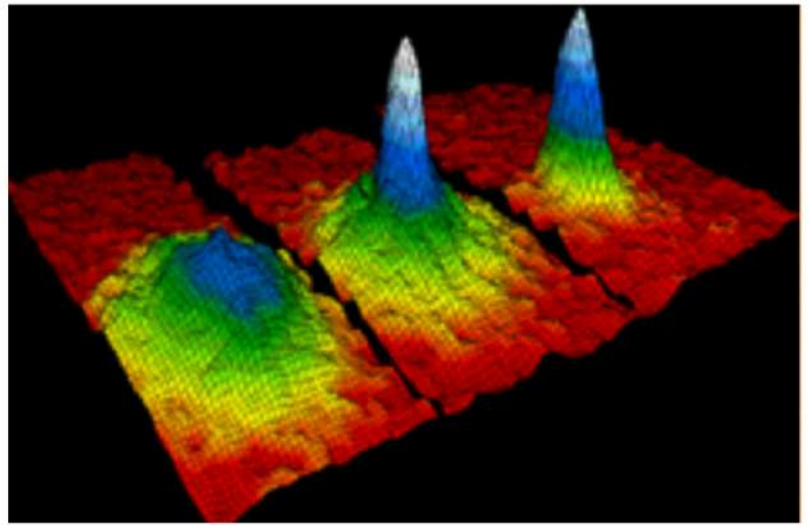
- It comprises over 99% of the visible universe.

- Plasma is often called "the fourth state of matter", along with solid, liquid and gas.



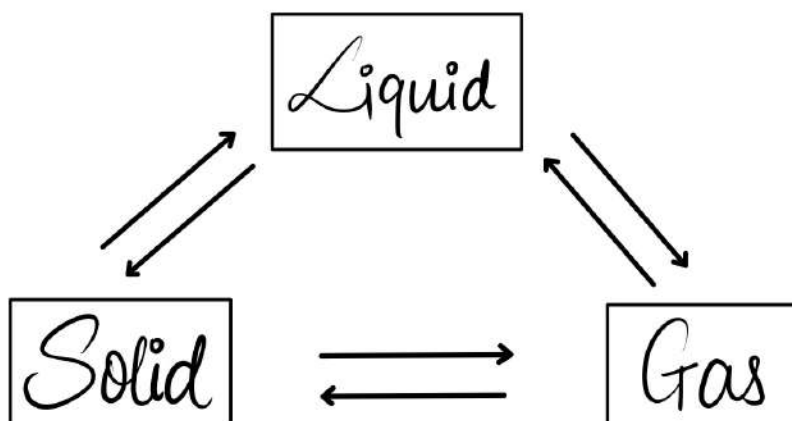
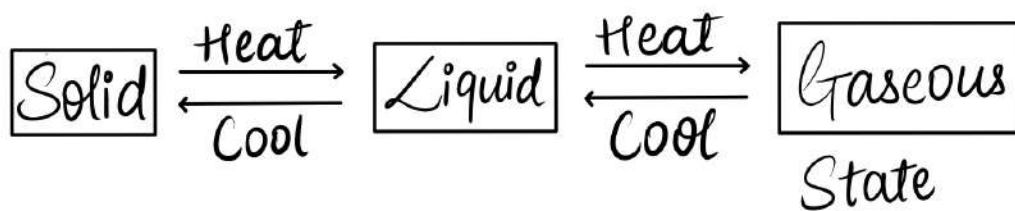
* BEC [Bose Einstein Condensate]

- In condensed matter physics, a Bose-Einstein condensate is a state of matter which is typically formed when a gas of bosons at low densities is cooled to temperatures very close to absolute zero.

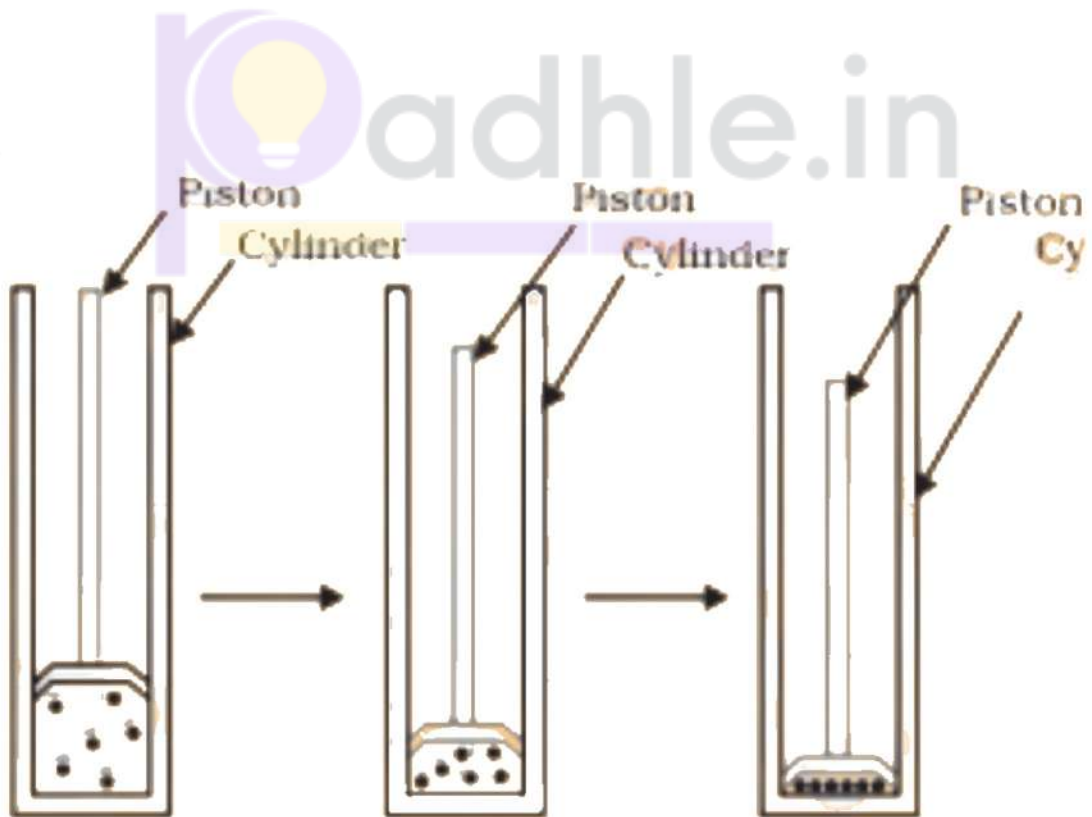
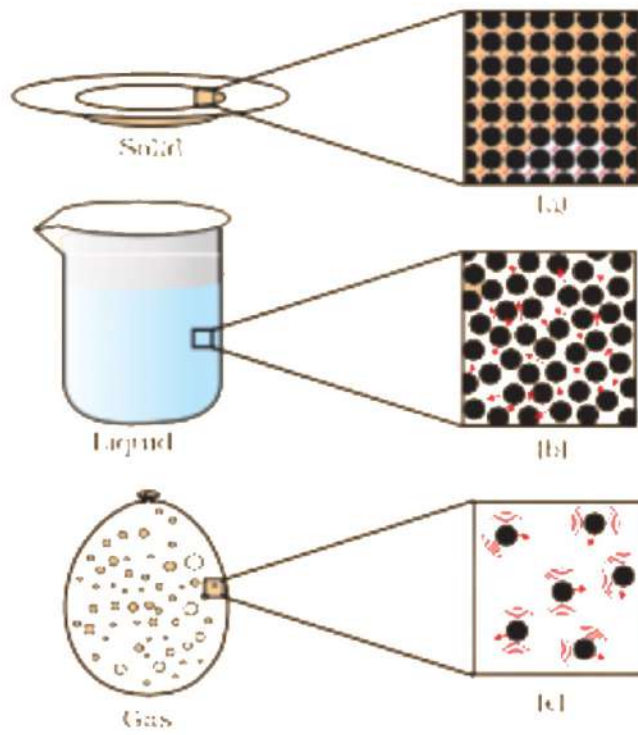


* What is Phase of Transformation?

→ Conversion of States :



Gas to Liquid using pressure



* Phase Transformation

- On increasing the temperature of solids, the kinetic energy of the particles increases.
- Due to the increase in kinetic energy, the particles start vibrating with greater speed.
- The energy supplied by heat overcomes the forces of attraction between the particles.
- The particles leave their fixed positions and start moving more freely.
- A stage is reached when the solid melts and is converted to a liquid.
- The minimum temperature at which solid melts to become a liquid at the atmospheric pressure is called its melting point.

- The melting point of a solid is an indication of the strength of the force of attraction between its particles.

* What is Melting Point?

- The temperature at which a substance will melt.

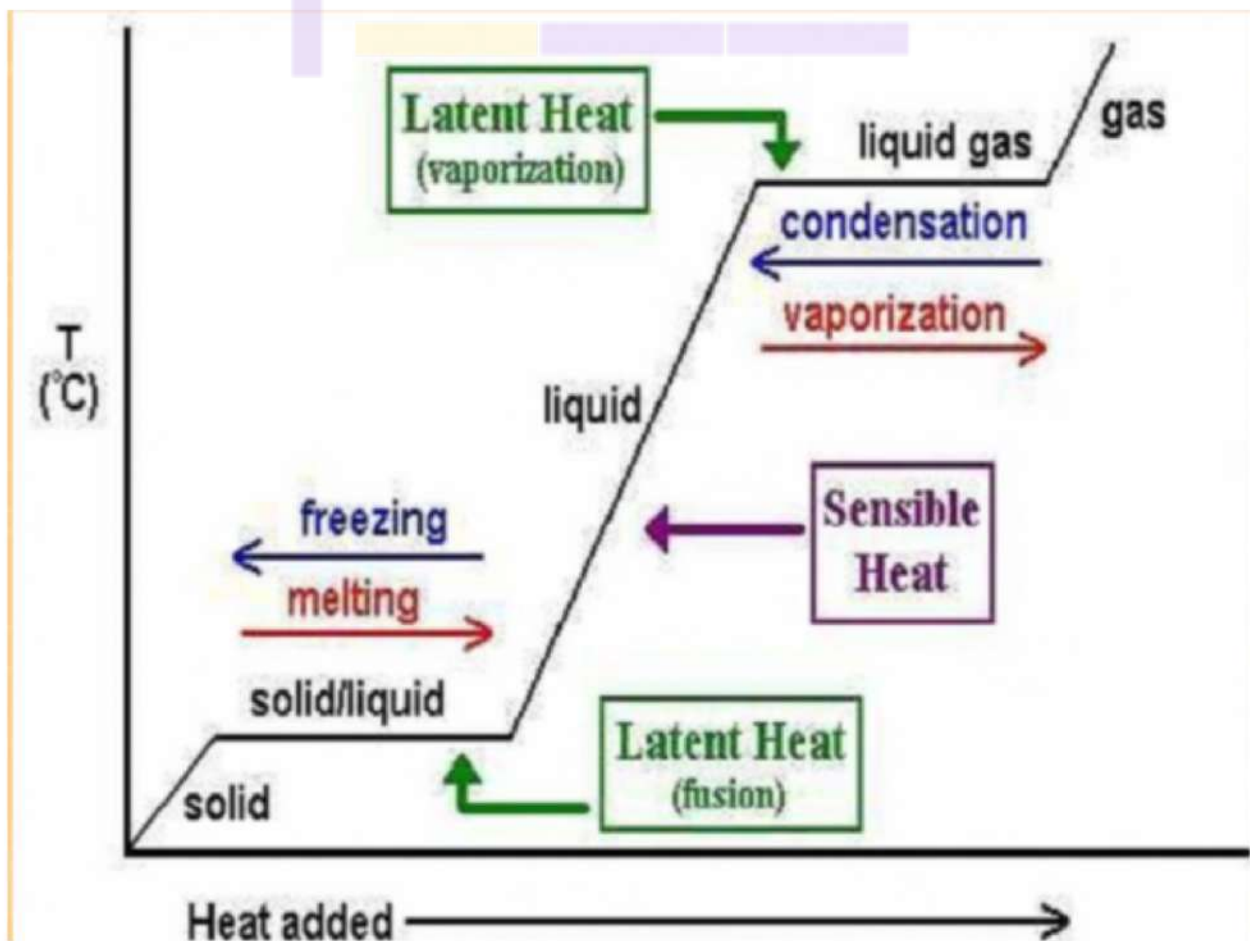
* What is Boiling Point ?

- The temperature at which a liquid starts to boil.

- During a phase change, the temperature remains constant.

- When a solid melts, its temperature remains the same, so where does the energy go?

* What is Latent Heat ?



* Latent Heat of Fusion :

- Amount of heat required to convert 1 kg of given solid into liquid.

* Latent Heat of Vaporisation :

- Amount of heat required to convert 1 kg of given liquid.

Unit - joule

* What is Evaporation?

- Evaporation is the process by which a liquid turns into a gas.
- It causes cooling.

<u>Evaporation</u>	<u>Boiling</u>
Takes place only on the surface.	Takes place throughout the liquid.
Liquid temperature drops during evaporation.	Liquid temperature remains unchanged during boiling.
Evaporation is a gradual process.	Boiling is a vigorous process, with the formation of bubbles throughout the liquid.

* Factors affecting Evaporation :

1. Temperature :

- Higher temperature of liquid, higher rate as more molecules at the surface are energetic enough to escape.

2. Humidity :

- Higher humidity, lower the rate as more water vapour present in the air to hit the escaped molecule back to liquid.

3. Boiling Point :

- Lower boiling point of liquid, faster rate as bonds between liquid molecules are weaker.

4. Surface Area of Liquid :

- Larger surface area, more molecules can escape from liquid at greater rate.

5. Movement of Air :

- Removes the liquid molecules as soon as they escape.

6. Pressure :

- Reducing atmospheric pressure increases rate of evaporation.

* What are Temperature Scales ?

$^{\circ}\text{C}$ to $^{\circ}\text{K}$

$$T_k = T_c + 273$$

$^{\circ}\text{C}$ to $^{\circ}\text{F}$

$$^{\circ}\text{F} = \frac{9}{5} ^{\circ}\text{C} + 32$$

